Deepak Mallampati

APPLIED AI ENGINEER - MULTIMODAL & DOCUMENT AI

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US-based, F-1 OPT (open to sponsorship) | Open to relocation

PROFESSIONAL SUMMARY

Applied AI engineer with 4+ years across LLMs, NLP, and multimodal AI, plus shipped, demoable products (MangaLens live Chrome extension; career-ops autonomous pipeline built with Claude Code). Strong on RAG, grounding, and schema-validated extraction, with multimodal/document experience from a production video-understanding pipeline (speech-to-text + transformer summarization) and computer-vision work (object detection). Comfortable prototyping customer POCs fast with Python and Claude CLI and deploying across AWS/containers.

SKILLS

Multimodal & Document AI • LLMs & NLP • RAG & Grounding • Computer Vision (Object Detection) • Rapid POC/Demo Delivery • Claude CLI • Cloud & VPC Deployment • Schema Extraction

Multimodal & CV: Video understanding (speech-to-text + transformer summarization) | Object detection | PyTorch, TensorFlow, Keras | OCR-adjacent extraction

LLMs & NLP: GPT-4o, Claude Sonnet, Gemini | RAG | Grounding & schema validation | NER, Text Classification, Summarization | LangChain/LangGraph | Fine-tuning (QLoRA)

Retrieval & Evaluation: FAISS, Pinecone, Weaviate | Cross-Encoder Re-ranking | RAGAS, BERTScore

Cloud & Delivery: Python, Claude CLI | AWS, Azure, GCP | Docker, Kubernetes, VPC | Spark/Databricks | Cross-functional stakeholder delivery

PROFESSIONAL EXPERIENCE

Progress Solutions *Jul 2025 - Present*

AI Engineer

USA

- Architected production-grade **agentic LLM systems** on a conductor-subagent orchestration framework, decomposing complex objectives into context-scoped subagents - reducing token consumption by **42%** while sustaining task accuracy across multi-step workflows.
- Engineered **retrieval-augmented generation (RAG)** pipelines over large-scale healthcare corpora, combining dense vector retrieval with cross-encoder re-ranking to lift answer relevance and materially reduce hallucination.
- Established a structured **LLM evaluation and monitoring framework** (RAGAS, BERTScore, custom metrics) with latency/accuracy dashboards to benchmark iterations and surface regressions before release.
- Optimized inference cost and latency through prompt compression, semantic caching, and model routing, sustaining SLA targets while reducing per-query overhead.
- Built fault-tolerant automation infrastructure with scheduled batch orchestration and exponential-backoff retry logic, reducing failed production runs by **60%**.

Vanguard *Jan 2024 - Jan 2025*

AI Engineer Intern

USA

- Developed **AI agents and backend services** for Vanguard's internal AI platform across **25+ AI agents** and workflows, integrating with production data pipelines and observability tooling.
- Optimized prompts and evaluated **GPT-4o, Claude Sonnet, and Gemini**, improving task success rates by **27%** and informing platform model-selection decisions.
- Built **12 APIs and microservices**, reducing p95 latency from 1.5s to **800ms** and supporting **100,000+ requests daily** across 20+ internal teams.
- Engineered safeguards and content controls for policy-compliant AI responses, reducing unsafe outputs by **42%** while maintaining quality.

Kent State University *Oct 2023 - Jan 2024*

ML & Gen AI Research Assistant

USA

- Fine-tuned **Qwen3 via 4-bit QLoRA** on an NVIDIA H100 using a curated instruction dataset, conditioning on a six-stage narrative schema for controllable long-form generation.
- Built a **privacy-preserving ML pipeline** (k-anonymity, l-diversity, Laplace differential privacy), cutting re-identification risk from **3.38% to 0.32%** while retaining 99% of baseline predictive performance.
- Authored a peer-review-ready IEEE-format conference paper documenting methodology and results.

Emerson *Jun 2022 - Apr 2023*

ML Trainee

India

- Built supervised regression/classification models for equipment-failure prediction, anomaly detection, and capacity forecasting on operational data.
- Fine-tuned **BERT and RoBERTa** for entity extraction and text classification, achieving **89% F1** on custom NER tasks.

PROJECTS

MangaLens Live on Chrome Web Store

Real-time AI translation of manga/webtoons across 29+ sites - a single inline action replacing a multi-step manual chore.

career-ops

Autonomous AI-driven job-search pipeline (built with Claude Code) that scans listings, scores fit, and generates tailored applications overnight without supervision.

Privacy-Preserving Clinical ML Pipeline

k-anonymity + l-diversity + differential privacy framework quantifying the privacy-utility tradeoff on sensitive patient data.

EDUCATION

M.S. in Computer Science - *Kent State University, OH* *2025*

GPA: 3.70 / 4.0